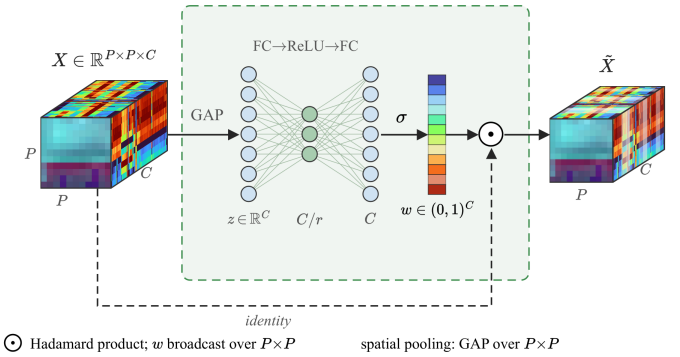
MODULE 1: SPECTRAL BAND ATTENTION MODULE (SBAM)

Network Stage: Stage I — Input Recalibration (Pre-Convolution) | **Parameter Footprint:** 10,225 params (3.32% of SGSCF-Net)



1. PROBLEM STATEMENT & ARCHITECTURAL MOTIVATION

- **Atmospheric Noise & Degradation:** Airborne hyperspectral sensors (AVIRIS/ROSIS) capture 100–200+ contiguous bands. Critical regions suffer from severe **atmospheric water vapor absorption** (bands 104–108 at 1.4 µm, bands 150–163 at 1.9 µm). If raw cubes enter 3D convolutions directly, noisy zero-transmission bands corrupt representations, propagating errors into deep layers.
- **Pre-Convolution Solution:** Unlike SENet or CBAM that operate strictly on deep latent feature maps, SBAM operates **directly on raw spectral bands before any convolution occurs**. It dynamically learns continuous soft attention weights per scene, shielding downstream representation learning.

2. MATHEMATICAL FORMULATION & TENSOR TRANSFORMATIONS

- **Step 1 (Spatial Compression):** For volumetric patch X ∈ ■^(B×1×P×P×C), adaptive 3D average pooling compresses spatial grid P×P to extract global spectral energy descriptor z ∈ ■^(B×C):
  z_c = (1 / P²) ∑_{u=1}^P ∑_{v=1}^P X(b, 1, u, v, c), for each band c ∈ {1, ..., C}.
- **Step 2 (MLP Bottleneck with Reduction r=8):** z passes through two linear projections with reduction ratio r=8 (bottleneck channels C_mid = max(■C/8■, 8)), applying ReLU δ and Sigmoid σ:
  w = σ(W■ · δ(W■ z + b■) + b■) ∈ (0, 1)^(B×C), where W■ ∈ ■^(C_mid×C) and W■ ∈ ■^(C×C_mid).
- **Step 3 (Channel-Wise Hadamard Modulation):** Attention vector w is broadcast across spatial dimensions ■(w) and multiplied element-wise with input cube X:
  X■ = X ■ ■(w) ∈ ■^(B×1×P×P×C). Noisy water absorption bands receive near-zero weights and are attenuated at the physical boundary.

3. SOTA BENCHMARK CONTRAST & VIVA DEFENSE SHIELD

- **What Was in Previous SOTA Models:** Existing models (e.g., SSRN, A2S2K-ResNet) either blindly process raw noisy bands through 3D convolutions, or apply channel attention late on deep intermediate layers after noise has already contaminated feature representations.
- **What We Did Better:** SBAM performs channel gating **upfront at the raw physical input**, learning soft continuous weights per scene that eliminate water-noise propagation before convolution with only 10,225 parameters.

Examiner: Why not simply delete noisy water absorption bands manually during data preprocessing?

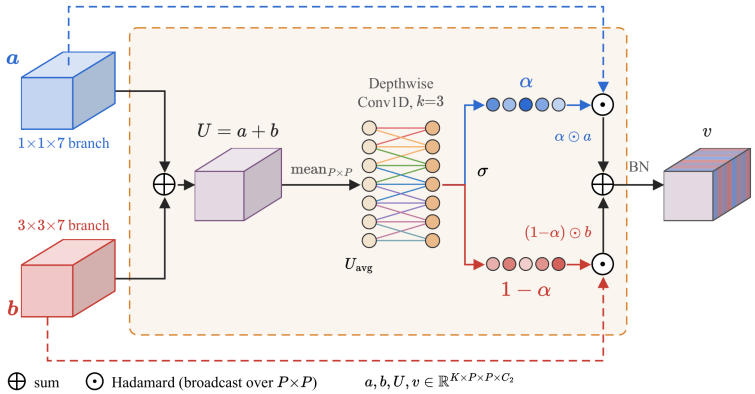
Winning Defense Answer: Manual band removal uses rigid, hardcoded indices (e.g. deleting 20 bands in AVIRIS). However, atmospheric attenuation and sensor noise vary across flight lines and times. SBAM dynamically learns continuous, data-driven attention weights per scene, preserving useful residual spectral signals adaptively.

4. WHAT TO UTTER IN YOUR PPT (Simple Words Spoken Script)

- **Exact Words to Speak:** *"Respected committee, let me explain **SBAM** in simple terms: Satellite hyperspectral images contain noisy atmospheric water absorption bands. **In previous SOTA models**, all bands were passed blindly into 3D convolutions, which spread noise into deep layers. **What we did better** is that BEFORE any convolution starts, SBAM pools the spatial patch into a per-band energy vector, runs it through an 8× reduction bottleneck, and produces an attention weight between 0 and 1. Corrupted bands get near-zero weights and are suppressed upfront through element-wise multiplication. It cleans the data right at the physical input layer using only 10,225 parameters (3.3% of the network)."*

MODULE 2: COMPLEMENTARY BRANCH SPECTRAL ATTENTION (CBSA)

Network Stage: Stage I — Dual-Stream Spectral Feature Entry | **Parameter Footprint:** 80 params (0.03% of SGSCF-Net)



1. PROBLEM STATEMENT & ARCHITECTURAL MOTIVATION

- **Multi-Branch Redundancy Problem:** Dual-stream 3D networks (e.g. Double-Branch CNNs, A2S2K-ResNet) extract features using parallel branches. However, combining them via direct summation (a + b) or shared attention assigns identical importance to both streams, causing co-dependent, duplicated feature representations and severe parameter explosion.
- **Complementary Branch Solution:** CBSA decouples spectral and spatial-spectral modeling into two dedicated parallel 3D streams, fused via a learned gating scalar α strictly constrained by **α + (1 - α) ≡ 1**. This forces orthogonal specialization and zero branch redundancy.

2. MATHEMATICAL FORMULATION & TENSOR TRANSFORMATIONS

- **Step 1 (Dual Receptive Fields):** Recalibrated patch X■ enters two parallel 3D convolutions (K=16 filters, spectral stride=2, C■ = ■(C-7)/2■ + 1):
  Branch a (Pure Spectral): a = ReLU(BN(Conv3D_{(1×1×7)}(X■))) ∈ ■^(B×16×P×P×C■) (captures narrow inter-band absorption lines).
  Branch b (Spatial-Spectral): b = ReLU(BN(Conv3D_{(3×3×7)}(X■))) ∈ ■^(B×16×P×P×C■) (captures joint spatial context).
- **Step 2 (Spatial Pooling & Depthwise Conv1D Gate):** Spatial context is pooled over fused features U = a + b:
  U_avg = (1 / P²) ∑_{u,v} (a + b)[:, :, u, v, :] ∈ ■^(B×16×C■).
  A depthwise 1D convolution (kernel k_s=3, grouped across 16 channels, only 48 params) models cross-band dependencies:
  α = σ(Conv1D_{k_s}(U_avg)) ∈ ■^(B×16×C■).
- **Step 3 (Complementary Convex Fusion):** The streams are fused via complementary combination and 3D batch normalization:
  v = BN(α ■ a + (1 - α) ■ b) ∈ ■^(B×16×P×P×C■).
  Because α + (1 - α) ≡ 1, the branches compete for capacity: when spectral fidelity dominates, α → 1; when spatial texture dominates, (1 - α) → 1.

3. SOTA BENCHMARK CONTRAST & VIVA DEFENSE SHIELD

- **What Was in Previous SOTA Models:** Prior dual-branch architectures (like Double-Branch CNN and CSCANet) fused parallel branches with simple element-wise addition or unconstrained independent gates, allowing both branches to learn identical, redundant representations.
- **What We Did Better:** CBSA enforces the strict partition of unity $\alpha + (1-\alpha) \equiv 1$. This mathematically prevents feature overlap and forces orthogonal specialization between pure spectral transitions and spatial context using only 80 parameters.

Examiner: Why enforce α and (1 - α) instead of learning two independent attention gates α and β?

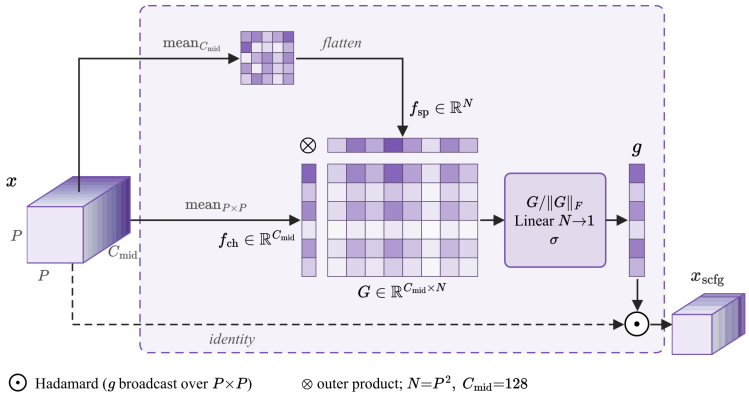
Winning Defense Answer: If α and β are learned independently, both can saturate to 1, causing high feature redundancy and doubling the feature magnitude. Enforcing α + (1 - α) = 1 mathematically guarantees that the branches compete and specialize into non-overlapping representations without duplicating parameters.

4. WHAT TO UTTER IN YOUR PPT (Simple Words Spoken Script)

- **Exact Words to Speak:** *"Moving to our second module, **CBSA** in simple terms: **Previous SOTA dual-branch models** combined narrow and wide 3D branches with simple addition, which caused both branches to learn the exact same duplicate features. **What we did better** is decoupling them into Branch a (1×1×7) for pure spectral absorption and Branch b (3×3×7) for spatial context, fused via α · **Branch a + (1 - α) · Branch b**. Because α + (1 - α) always sums to 1, the branches are mathematically forced to specialize: when spectral fidelity is needed, α goes to 1; when spatial context dominates, (1 - α) takes over. Branch redundancy is mathematically zero, costing only 80 parameters."*

MODULE 3: SECOND-ORDER COVARIANCE FUSION GATE (SCFG)

Network Stage: Stage I ■■■ Stage II Transition Bridge (Spectral to Spatial) | **Parameter Footprint:** 82 params (0.03% of SGSCF-Net)



1. PROBLEM STATEMENT & ARCHITECTURAL MOTIVATION

- **The Flaw of First-Order GAP:** Standard spatial attention mechanisms and global pooling compute only the **first-order spatial mean** μ_c = (1/N) ∑ x_{c,i}. In remote sensing, this discards spatial-channel co-activations (e.g. how specific spectral channels correlate with edges vs homogeneous interiors), failing to separate spectrally similar materials like asphalt and gravel, or subtle shadow gradients.
- **Second-Order Covariance Solution:** SCFG computes a **bilinear outer product G = f_ch ⊗ f_sp** between the channel descriptor and spatial position descriptor, normalized via the Frobenius norm. This models second-order statistics with zero iterative decompositions.

2. MATHEMATICAL FORMULATION & TENSOR TRANSFORMATIONS

- **Step 1 (Spectral Collapse & Channel Expansion):** Adaptive 3D average pooling collapses spectral dimension C■ → 1; Conv3D_{(1×1×1)} expands channels 16 → C_mid = 128. Squeezing yields 2D spatial feature map x ∈ ■^(B×128×P×P) with N = P × P pixels.
- **Step 2 (Dual Descriptors):** Global channel descriptor f_ch and spatial position descriptor f_sp are extracted:
  f_ch(b, c) = (1 / N) ∑_{n=1}^N x(b, c, n) ∈ ■^(B×128), **f_sp(b, n) = (1 / 128) ∑_{c=1}^{128} x(b, c, n) ∈ ■^(B×N)**.
- **Step 3 (Bilinear Covariance Matrix via Outer Product):** Outer product captures cross-domain co-occurrences:
  G = f_ch ⊗ f_sp = f_ch f_sp^T ∈ ■^(B×128×N), where G(b, c, n) = f_ch(b, c) · f_sp(b, n).
- **Step 4 (Frobenius Normalization & Gating):** Normalized to stabilize gradient backpropagation without matrix square roots:
  G■ = G / (■G■_F + ε), where ■G■_F = √(∑_c ∑_n G(b, c, n)²) and ε = 10■■.
  Linear projection W_G ∈ ■^(1×N) produces channel gate: **g = σ(W_G G■) ∈ ■^(B×128)**.
  Spatial features are modulated element-wise: **x_scfg = x ■ g**.

3. SOTA BENCHMARK CONTRAST & VIVA DEFENSE SHIELD

- **What Was in Previous SOTA Models:** Standard architectures (SSRN, SSAN, ViT) rely entirely on first-order Global Average Pooling (GAP), which only measures average intensity per channel and completely discards spatial-channel covariance structures.
- **What We Did Better:** SCFG computes a bilinear outer product $G = f_{\text{ch}} \otimes f_{\text{sp}}$ normalized by Frobenius norm, capturing second-order cross-domain correlations without expensive matrix square roots using only 82 parameters.

Examiner: What physical information does the outer product f_ch ⊗ f_sp capture in remote sensing?

Winning Defense Answer: In remote sensing, pixel reflectance is strongly coupled to spatial context. First-order pooling averages them out independently. The outer product captures second-order co-occurrence: how specific spectral absorption bands vary across spatial coordinates. This lifted shadow classification to 99.34% and gravel accuracy to 99.76% on Pavia University.

4. WHAT TO UTTER IN YOUR PPT (Simple Words Spoken Script)

- **Exact Words to Speak:** *"Finally, our third module, **SCFG** in simple terms: **Previous SOTA models** relied on simple Global Average Pooling, which only measures the average intensity of each channel and misses how spectral channels co-vary across space—causing confusion between asphalt and gravel. **What we did better** is computing the bilinear outer product **G = f_ch ⊗ f_sp** to capture second-order spatial-channel covariance. We normalize it with the Frobenius norm so gradients stay stable, and project it into a spatial gate. This captures subtle material boundaries using only 82 parameters and lifted shadow accuracy to 99.34%."*